



**AMERICAN SOCIETY  
FOR RADIATION ONCOLOGY**

251 18<sup>th</sup> St. South, 8<sup>th</sup> Floor  
Arlington, VA 22202

Main: 703.502.1550

Fax: 703.502.7852

[www.astro.org](http://www.astro.org) · [www.ranswers.org](http://www.ranswers.org)

---

Board of Directors

**CHAIR**

**Sameer Keole, MD, FASTRO**

Mayo Clinic  
Phoenix

**PRESIDENT**

**Neha Vapiwala, MD, FASTRO**

University of Pennsylvania  
Philadelphia

**PRESIDENT-ELECT**

**Catheryn Yashar, MD, FASTRO**

University of California San Diego  
San Diego

**IMMEDIATE PAST CHAIR**

**Howard Sandler, MD, MS, FASTRO**

Cedars-Sinai Medical Center  
Los Angeles

February 23, 2026

Robert F. Kennedy, Jr.

Department of Health and Human Services

ATTN: HHS Health Sector AI RFI

Mail Stop: 7033A, 330 C Street SW

Washington, DC 20201

Submitted electronically: <http://www.regulations.gov>

Dear Secretary Kennedy:

The American Society for Radiation Oncology (ASTRO)<sup>1</sup> appreciates the opportunity to provide written comments on the Department of Health and Human Services' (HHS) Health Sector Artificial Intelligence (AI) request for information (RFI). The Department seeks feedback on the actions it can take to establish a forward-leaning, industry-supportive, and secure approach to accelerate the adoption and use of AI as part of clinical care. In general, HHS seeks feedback on ways in which these approaches can be most effectively applied to support the rapid adoption and use of AI in clinical care, to foster public trust and confidence in modern technology solutions, to reduce uncertainty that impedes AI innovation, and to align federal incentives so that AI is deployed in ways that enhance productivity, reduce burden, lower health care costs, and improve health outcomes for patients, caregivers, and communities.

AI has significant potential to improve patient outcomes, enhance access to care, and increase health system efficiency. However, successful integration into clinical care requires:

- Physician-guided implementation to ensure patient safety and compassionate care;
- Thoughtful reimbursement models tied to measurable value;
- Infrastructure investment incentives; and
- Medical specialty society leadership in accreditation and appropriate use frameworks.

---

<sup>1</sup> ASTRO members are medical professionals practicing at hospitals and cancer treatment centers in the United States and around the globe. They make up the radiation treatment teams that are critical in the fight against cancer. These teams include radiation oncologists, medical physicists, medical dosimetrists, radiation therapists, oncology nurses, nutritionists, and social workers. They treat more than one million patients with cancer each year. We believe this multi-disciplinary membership makes us uniquely qualified to provide input on the inherently complex issues related to Medicare payment policy and coding for radiation oncology services.

AI should augment—not replace—clinical judgment. Federal policy should ensure that AI deployment strengthens human-centered medicine while creating sustainable pathways for responsible adoption. HHS payment policy has a tremendous influence on whether high-value AI innovations are adopted in clinical care, yet the current payment systems are poorly suited to support tools whose value accrues over time through improved safety, efficiency, and outcomes. To ensure that the promise of AI is not diminished by the current payment landscape, HHS should modernize payment policy to better align incentives with long-term value. This includes creating pathways that reward AI tools demonstrating measurable improvements in patient outcomes, safety, access, or workforce sustainability, rather than tying payment solely to discrete services or assumed efficiency gains. Aligning payment policy with value and real-world implementation costs will be essential to accelerating access to effective and affordable AI tools in clinical care.

Additionally, ASTRO is concerned about the rapid adoption of AI by health insurers, particularly in areas such as utilization management, prior authorization, and claims adjudication.<sup>2</sup> Unlike AI tools used by clinicians, which are subject to FDA oversight and professional accountability, insurer-deployed AI operates with far less transparency or standardized review. This lack of oversight raises concerns about how decisions are made, how errors are identified, and how patients and providers can meaningfully challenge adverse determinations.

ASTRO is concerned that deploying AI into already contentious processes (e.g., prior authorization and coverage determinations) may simply accelerate inappropriate or poorly explained denials, rather than improve decision quality. Faster decisions are only beneficial if they are accurate, clinically grounded, and based on transparent criteria applied in good faith. Without clear disclosure of when AI is used, what data it relies on, and how its outputs are monitored for bias or error, trust between payers, providers, and patients is likely to erode. As HHS considers modernizing payment systems to support high-value AI, it is essential that similar expectations for transparency, accountability, and ongoing monitoring apply to AI used by payers, particularly when those tools influence access to medically necessary care.

Moreover, it is critical that any funding allocated by HHS for AI initiatives in clinical care comes from new appropriations, rather than diverting resources from existing Medicare Part B funds. Drawing from current Medicare budgets could jeopardize essential services and undermine the financial stability of the program. To ensure that AI adoption enhances care without compromising established coverage, HHS should seek dedicated, supplemental funding streams specifically earmarked for AI infrastructure, evaluation, and implementation support. This approach will protect Medicare beneficiaries while fostering innovation and sustainable progress in health care delivery.

### **Specific Questions**

In the RFI, HHS posed 10 specific questions. ASTRO's responses to each are below.

#### **1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?**

---

<sup>2</sup> *In a financial pinch, major health insurers are turning to AI for help,*  
<https://www.statnews.com/2026/02/17/health-insurers-expand-ai-use-undermining-providers-trust/>

Reimbursement and the absence of viable payment pathways are the dominant barriers. Radiation oncology practices have limited financial margins to absorb new AI tools, and these AI tools often lack explicit coding and payment recognition. This makes widespread AI adoption financially infeasible even when clinically beneficial.

Additionally, interoperability and data fragmentation remain major headwinds. Siloed electronic health record (EHR) systems, limited access to structured data, and inconsistent standards implementation restrict development, evaluation, and scale. In radiation oncology specifically, interoperability must include the physics data chain (e.g., contour geometry, dose grids, beam parameters), not just imaging; otherwise, it becomes ineffective and potentially unsafe (e.g., re-irradiation planning without access to prior dose grids).

Finally, AI “maturity” varies widely. Some use cases are mature and in broad clinical use, while others, such as generative AI for high-stakes clinical tasks, are not reliable and may hallucinate or fail domain reasoning. Policy that treats all AI as equivalent can block safe adoption of mature tools while under-addressing risks in immature categories.

**2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care?**

First, reimbursement for AI-enabled clinical work should not assume that AI reduces physician work. ASTRO members regularly provide feedback that AI often actually *increases* work (e.g., to commissioning, quality assurance, or monitoring), rather than reducing or eliminating it. Moreover, it is essential to track and disclose how financial resources allocated for AI adoption are distributed: whether they benefit the practice, the provider, or the AI company. This will ensure transparency, accountability, and equitable support for clinical care innovation.

Next, HHS should issue clear national guidance on AI governance and accountability within health systems, particularly for post-deployment use. Today, both clinicians and institutions face uncertainty about expectations for oversight, documentation, monitoring, and responsibility allocation, especially in multi-vendor environments where AI tools interact with EHRs, imaging systems, treatment planning systems, and delivery devices. This uncertainty slows adoption and increases risk. HHS can play a critical role by articulating baseline governance expectations applicable to both FDA-regulated AI devices and non-device clinical AI. Such guidance should clarify that physicians remain the final decision-makers in clinical care, while also establishing expectations for transparent logging of AI recommendations, continuous performance monitoring, version and update management, and incident reporting.

Additionally, HHS should actively incentivize and support prospective evaluation of AI tools, both before and after deployment. ASTRO members consistently emphasize that the lack of standardized, prospective evaluation frameworks undermines confidence in AI and slows responsible adoption. Pre-deployment expectations should include external validation across institutions, robustness testing, and subgroup fairness analysis. Post-deployment, evaluation must extend beyond static accuracy metrics to include drift monitoring, real-world performance dashboards, and outcomes-based surveillance. HHS can accelerate this work by supporting benchmarking consortia that maintain shared evaluation datasets, develop common metrics, and enable cross-institution validation. Federal support through grants, cooperative agreements, or similar mechanisms would help create the infrastructure needed for continuous learning while reducing duplicative local effort.

Finally, HHS should provide clarity that routine local AI validation, monitoring, and safety surveillance, when conducted with appropriate safeguards, are part of normal health care operations and not automatically classified as research. This distinction is critical for enabling continuous safety improvement, particularly in community-based settings.

**3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?**

The increasing use of non-medical device AI in clinical care raises a set of legal and implementation challenges that are not well addressed by existing governance and accountability structures. Unlike FDA-regulated medical devices, many AI tools used in clinical environments (e.g., clinical documentation tools, prior authorization support), can have direct and material effects on patient access, timeliness of care, and clinical decision-making without being subject to a clear, uniform oversight framework.

One of the most significant challenges is uncertainty around accountability and liability. Health systems and clinicians often lack clear guidance on how responsibility should be allocated among clinicians, health systems, and vendors when non-device AI tools influence clinical workflows or recommendations. Federal guidance clarifying that physicians remain the final decisionmakers in clinical care is needed, while also establishing clear expectations for how AI outputs are reviewed, documented, and acted upon. Without this clarity, organizations may either over-rely on AI outputs or, instead, delay adoption of beneficial tools due to liability concerns.

Another issue is the lack of standardized transparency and auditability requirements. Many clinical AI tools function as “black boxes,” providing recommendations or workflow actions without sufficient visibility into how outputs were generated. This creates risks for patient safety, compliance, and trust. ASTRO supports requirements for transparent logging of AI recommendations, including the ability to retrospectively review when and how AI influenced clinical workflows. Such logging is essential not only for accountability and quality improvement, but also for incident investigation and patient safety surveillance.

**4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?**

Effective evaluation of non-medical device AI in clinical care requires approaches that reflect real-world clinical use and patient safety. Prior to deployment, the most promising evaluation methods include external validation across multiple institutions, robustness testing under realistic clinical conditions, and subgroup analyses to assess performance across patient populations. These steps are critical for identifying risks that may not be apparent in development datasets and for building clinician confidence in AI tools.

Post-deployment evaluation is equally important and should be treated as a core safety expectation. AI tools are frequently updated or retrained over time, creating ongoing risk of performance drift. Continuous monitoring, supported by transparent dashboards and audit logs, is therefore essential.

Importantly, evaluation should emphasize outcomes-based and workflow-level impacts, such as safety events, care delays, or access implications, rather than relying solely on technical accuracy metrics.

HHS can play an important role by supporting shared evaluation infrastructure rather than leaving each health system to develop duplicative approaches. ASTRO would support federal investment in benchmarking consortia that maintain shared evaluation datasets, define common metrics, and enable external validation across institutions.

**5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?**

HHS can best support private-sector activities by providing clear expectations that promote safety, transparency, and sustainability without constraining innovation. Successful AI adoption depends on predictable frameworks that recognize the full clinical and operational lifecycle of AI, rather than focusing narrowly on algorithm performance or assumed efficiency gains.

As previously mentioned, HHS should ensure that payment and program policy do not discourage participation in accreditation or certification efforts by assuming AI reduces clinical work and improves quality. Recognizing this reality will help ensure that private-sector certification and accreditation programs remain viable across diverse practice settings, including community-based care.

Second, HHS should promote standardized transparency and documentation expectations for clinical AI tools. Clear, common expectations around intended use, limitations, performance variability, update management, audit logging, and cybersecurity would reduce duplication, improve comparability, and enable more meaningful industry-driven testing and credentialing.

Finally, HHS should support voluntary, consensus-based accreditation models that focus on institutional AI governance and readiness, as well as independent third-party benchmarking efforts that enable external validation across institutions. Federal support for shared evaluation infrastructure and AI literacy standards would amplify private-sector leadership while ensuring that AI adoption advances patient safety and trust.

**6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?**

AI tools deployed in clinical care have shown the greatest success when applied to well-defined, mature use cases that augment existing workflows rather than attempting to replace complex clinical judgment. AI has met expectations in some areas, such as document summarization, but has fallen short of expectations when applied to high-stakes clinical decision-making. For example, generative AI systems, may hallucinate clinical facts, fabricate citations, or fail to apply core domain reasoning, making them unreliable for tasks such as treatment intent formulation or dose specification. In these settings, the gap between technical performance and clinical reliability remains significant, and premature deployment can introduce safety risks rather than reduce them.

Looking ahead, the most promising AI tools are those that directly address persistent challenges in healthcare delivery, including treatment delays, staffing shortages, and unwarranted variability in care. The greatest value will come from tools that provide new insights into quality and safety across the full care pathway, rather than narrowly optimizing isolated tasks. When paired with appropriate reimbursement, governance, and evaluation frameworks, these applications offer meaningful opportunities to improve care while helping control long-term costs.

**7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?**

Within health care organizations, adoption of AI for clinical care is most strongly influenced by those responsible for quality, safety, and operations. While executive and IT leadership play important roles, AI adoption ultimately depends on whether tools align with clinical workflows, patient safety priorities, and workforce realities. In specialties such as radiation oncology, decisions are heavily shaped by radiation oncologists, physicists, and multidisciplinary oversight committees responsible for validation, quality assurance, and ongoing monitoring.

The primary administrative hurdles to adoption include lack of sustainable reimbursement, unclear governance and accountability expectations, and the operational burden associated with commissioning, validating, and monitoring AI tools over time. Uncertainty around post-deployment responsibilities, like performance drift monitoring, version control, and update management, further slows adoption. Staffing shortages compound these challenges, as organizations may lack the protected time and resources needed to deploy AI safely, even when the technology could help address access delays and workforce strain. Together, these barriers often limit adoption of clinically valuable AI tools despite strong interest from clinicians.

**8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.**

Current interoperability efforts are predominantly structured around diagnostic workflows, prioritizing the seamless exchange of imaging data for purposes such as diagnosis and initial assessment. While these initiatives have advanced the ability to share and interpret diagnostic images across institutions, they often fall short in supporting therapeutic specialties like radiation oncology. In these settings, images serve merely as the starting point to a much more intricate process that encompasses not just visualization, but also the computational planning and precise delivery of radiation. Therefore, AI development and safe clinical use in therapeutic domains require interoperable access that extends beyond images to include treatment-specific data such as contour geometry, dose distributions, beam parameters, and longitudinal treatment history. Without robust support for the full spectrum of therapeutic data, interoperability frameworks risk limiting the potential impact of AI-enabled care in these critical specialties.

In radiation oncology, the lack of standardized, computable exchange of prior treatment data remains a major barrier to AI-enabled care, research, and safety monitoring. For example, without interoperable access to prior dose information, AI tools cannot reliably support re-irradiation planning, toxicity

prediction, or adaptive treatment strategies. Expanding interoperability standards to include radiation treatment history and therapy-specific data elements would meaningfully widen opportunities for AI innovation while improving patient safety. By prioritizing interoperability that reflects real clinical workflows and specialty-specific data needs, HHS can help unlock the full potential of AI to improve quality, safety, and access across the healthcare system.

**9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?**

From ASTRO members' perspective, patients and caregivers are most interested in AI applications that reduce delays in care, improve safety, and support timely access to high-quality treatment and improved outcomes. At the same time, patients and caregivers have legitimate concerns about safety, transparency, and trust in AI-enabled clinical care. Tools that operate as "black boxes" produce unreliable or fabricated outputs, or are deployed without clear human oversight can undermine confidence and raise fears about errors affecting care decisions. Patients also expect that AI will augment, *not replace*, clinical judgment and that physicians remain accountable for decisions affecting their treatment.

Privacy and appropriate use of health data are additional concerns, particularly as AI tools rely on large volumes of clinical information and may evolve over time. Patients want assurance that their data is protected, that AI tools are monitored for ongoing performance, and that failures or biases are identified and addressed promptly.

**10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?**

HHS should prioritize research focused on real-world performance, safety, and sustainability, rather than algorithm development alone. A key priority is prospective, multi-institutional evaluation research that examines longitudinal performance, subgroup variability, and interaction between AI tools and human decision-making. Research should move beyond retrospective accuracy metrics to assess real-world outcomes such as patient safety, access to care, workflow reliability, and treatment delays. Shared datasets and benchmarking infrastructure would support this work and reduce duplicative effort across health systems.

HHS also should prioritize research on the human and organizational dimensions of AI adoption, including governance models, oversight structures, and best practices for commissioning, monitoring, and updating AI systems. Understanding how AI shifts clinical and administrative work (and the implications for staffing and safety) is essential to scalable adoption.

**a. Are there published findings about the impact of adopted AI tools and their use clinical care?**

The published literature on AI adoption in clinical care shows the strongest and most consistent evidence of benefit in narrow, well defined use cases, particularly those that automate repetitive tasks or support clinical prioritization rather than replacing complex clinical judgment. By contrast, the literature remains more limited and mixed for higher-stakes or more generative AI applications that influence clinical reasoning or treatment decisions.

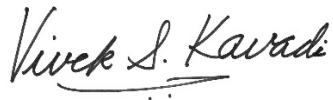
**b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?**

The literature increasingly recognizes that the costs and benefits of AI in clinical care are not evenly distributed and rarely result in immediate net savings. While AI may reduce time spent on certain tasks or improve efficiency, these benefits are often offset by new and ongoing tasks and costs related to data infrastructure, system integration, validation, quality assurance, cybersecurity, and continuous performance monitoring. As a result, AI frequently increases work and costs rather than eliminating them.

Many studies also note that financial benefits may accrue to payers or the broader health system, while implementation and oversight costs are borne by clinicians and health care organizations. In addition, downstream benefits such as improved safety, reduced delays, or avoided adverse events are difficult to quantify and are often underrepresented in traditional cost analyses. Overall, the literature suggests AI delivers the greatest value when payment and policy frameworks account for these cost transfers and support the investments required for safe, sustained use.

Thank you for the opportunity to comment on this RFI. We look forward to continued dialogue with HHS officials. Should you have any questions on the items addressed in these comments, please contact Adam Greathouse, Director of Health Policy, at (703) 839-7376 or [Adam.Greathouse@astro.org](mailto:Adam.Greathouse@astro.org).

Respectfully,



Vivek S Kavadi, MD, MBA, FASTRO  
Chief Executive Officer



Sameer R. Keole, MD, FASTRO  
Chair, ASTRO Board of Directors